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Mohammad Vazir Panah

Work permit: Italian **Nationality:** Iranian **Date of birth:** 23 Apr 1994

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ABOUT MYSELF

Research Interests: **Embedded Systems | Cyber-Physical Systems (CPS) | High-Performance Computing (HPC)**
Expertise in **FPGA** and **RISC-V architectures**
Passionate about building scalable and reliable systems across **heterogeneous CPU/GPU** platforms

EDUCATION AND TRAINING

Master of Science (M.Sc.) in Computer Engineering

Shahid Beheshti University [Sep 2019 – Oct 2022]

City: Tehran | **Country:** Iran | **Field(s) of study:** Computer Systems Architecture | **Final grade:** 3.79 / 4 |
Thesis: A framework for modeling self-aware cyber-physical systems with SystemC

Bachelor of Science (B.Sc.) in Computer Engineering

Birjand University [Sep 2012 – May 2016]

City: Birjand | **Country:** Iran | **Field(s) of study:** Hardware Engineering | **Final grade:** 3.40 / 4 | **Thesis:**
Simulation and Programming the Alpha Rex Human-Robot Produced by Lego Company

WORK EXPERIENCE

University of Salerno – Salerno, Italy

City: Salerno | **Country:** Italy | **Name of unit/department:** Department of Computer Science - ISISLab

Research Fellow

[Jan 2023 – Current]

- **Scientific Responsible: Prof. Biagio Cosenza**
- Focus on **High-Performance Computing** and **RISC-V** architectures
- Contributed to LibreRT project, developing and deploying real-time applications on embedded CPUs and GPUs
- Evaluating compiler performance on RISC-V vector boards
- Collaborating on research and co-authoring publications
- Deployed SoC boards; managed experiments with Git, Docker, CI; documented setups for reproducibility

Sepehr Educational Institutions – Kashmar, Iran

City: Kashmar | **Country:** Iran

Instructor

[2013 – 2015]

- Delivered theoretical and practical instruction in C++ programming
- Taught ICDL (International Computer Driving License) skills
- Taught CorelDRAW and Adobe FreeHand, focusing on vector graphics

Abooreyhan Company – Kashmar, Iran

City: Kashmar | **Country:** Iran

Education Intern

[2014 – 2015]

- Experienced in hardware/software troubleshooting
- Assisted in hardware repair and maintenance, including diagnostics and component replacement
- Assembled and configured PCs, installing hardware devices
- Software installation and system setup

ACADEMIC EXPERIENCES

Teaching Assistance

- **Supervisor: Dr. Seyed-Hosein Attarzadeh-Niaki**
- M.Sc. Courses | Shahid Beheshti University
  - Design and Analysis of Real-Time Embedded Systems | 2021
  - System-on-Chip | 2020, 2021
  - Advanced Computer Architecture | 2021
  - Fundamentals of Embedded and Real-Time Systems | 2020
- **Supervisor: Dr. Mohammad Hadi Valavi**
- B.Sc. Course | Birjand University
  - Computer Architecture | 2016

Scientific Laboratory Membership

- ISISLab laboratory | HPC Group | Salerno, Italy
- SBU Cyber-Physical Systems Lab Research Team | Tehran, Iran

PUBLICATIONS

[2025]

A Performance Analysis of Autovectorization on RVV RISC-V Boards

Authors: L. Carpentieri, M. VazirPanah, and B. Cosenza | **Journal Name**: PDP-33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing | **Publisher**: IEEE

[2023]

System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs

Authors: S.-H. Attarzadeh-Niaki and M. Vazirpanah | **Journal Name**: Journal of Innovations in Computer Science and Engineering (JICSE) | **Publisher**: Faculty of Computer Science and Engineering, Shahid Beheshti University

[2022]

ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design

Authors: M. Vazirpanah, S.-H. Attarzadeh-Niaki, and A. Salimi-Badr | **Journal Name**: 27th International Computer Conference, Computer Society of Iran (CSICC) | **Publisher**: IEEE

TECHNICAL SKILLS AND EXPERTISE

SKILLS

Programming: C, C++, Python, MATLAB, Assembly

Parallel Models: OpenMP, OpenCL, SYCL

Hardware/Embedded: FPGA, RISC-V, VHDL, Verilog, SystemC, FreeRTOS

Simulation & Tools: Gem5, QEMU, OVPsim, Gazebo, Quartus, ModelSim, ROS, MATLAB-Simulink, Ansys Scade Suite

Other Tools: Git, Docker, Linux, OpenModelica, NuSMV, UPPALL, LaTeX

PROJECTS

LibreRT

- LibreRT - Portable Heterogeneous Real-time Programming for the Embedded Computing Continuum
- Research project funded under the Italian PRIN 2022 program
- Role: development and deployment of real-time applications on embedded CPUs and GPUs
- Tools: SYCL, OpenMP, OpenCL, and RISC-V boards

RISC-V Benchmark Suite

- RISCV-bench - A repository that contains a set of benchmarks for RISC-V and RVV
- Compared various versions of modern compilers, such as GCC and LLVM

A framework for modeling self-aware cyber-physical systems with SystemC

- Designed a self-aware CPS model using ForSyDe-SystemC and ROS co-simulation
- Developed SADF model and contributed to the ForSyDe-SystemC

Implemented FIR Filter and Tiny YOLO1 on the DE0-Nano/DE10 Intel FPGA board

- Designed and implemented an IP core in Quartus for FIR filtering via Avalon MM interface
- Implemented an optimal and real-time implementation of the TinyYOLO1 object recognition system

HONOURS AND AWARDS

[2020] R&D Center of the Mobile Telecommunication Company of Iran (MCI)
Research Grant *Winner of the national R&D research grant competition among Master's and PhD students.*

[2016] Birjand University
Top Computer Engineering Students *Ranked in the top 15% of BSc Computer Engineering students*

LANGUAGE SKILLS

Mother tongue(s): Persian

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Italian

LISTENING A2 **READING** A2 **WRITING** A2

SPOKEN PRODUCTION A2 **SPOKEN INTERACTION** A2

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

REFERENCES

References

Research Supervisor

Prof. Biagio Cosenza | University of Salerno, Italy | Email: bcosenza@unisa.it

Head of ISISLab

Prof. Vittorio Scarano | University of Salerno, Italy | Email: vitsca@unisa.it

M.Sc. Supervisor

Dr. Seyed-Hosein Attarzadeh-Niaki | Shahid Beheshti University, Iran | Email: h_attarzadeh@sbu.ac.ir